Ex. No : 02 Date :

Study of different types of Network cables.

Aim :

To study different types of Network cables.

a) Understand different types of Network cable :

Different type of cables used in networking are :

i) Unshielded Twisted Pair (UTP) cable
ii) Shielded Twisted Pair (STP) cable
iii) Coaxial cable
iv) Fibre Optic cable.

Cable type	category	Maximum Data Transmission	Advantages / Disadvantages	Application / use.
UTP	category 3	10 bps	Advantages : i) cheaper in cost ii) easy to install as they have a smaller overall diameter. Disadvantages : i) More prone to (EMI) Electromagnetic interference & noise	i) 10 Base-T Ethernet
	category 5	upto 100 Mbps		ii) Fast Ethernet, Gigabit Ethernet.
	Category 5e	1 Gbps.		iii) Fast Ethernet, gigabit Ethernet.

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / use.
STP	category 6, 6a	10Gbps	Advantages: i. Shielded. ii. Faster than UTP iii. Less susceptible to noise & interference. Disadvantages: i. Expensive ii. Greater Installation effort	Gigabit Ethernet, 10G Ethernet (55m) Widely used in data centres.
SSTP	category 7	10 Gbps		Gigabit Ethernet, 10G Ethernet (100 m)
Coaxial Cable	RG-6 RG-59 RG-11	10 - 100 Mbps	Advantages: i. High bandwidth ii. Immune to interference. iii. Low loss bandwidth. iv. Versatile Disadvantages: i. cost ii. size is bulky.	Speed of signal is 500m Television network High speed Internet connection
fibre optics cable	single mode Multi mode	100 Gbps	Advantages: i. High speed. ii. High bandwidth iii. High security iv. Long distance. Disadvantages: i. Expensive ii. Requires skilled Installers	Maximum distance of fibre optics Cable is around 100 meters.

b) Make your own Ethernet Cross-Over cable / Straight cable :

Tools & Parts needed :

i) Ethernet cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.

ii) A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug & strip & cut the shielding off the cables.

iii) Two RJ45 plugs.

iv) Optional two plug shields.

Step 1 : To start construction of the device, begin by threading shields onto the cable.

Step 2 : Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3 : After, you will need to untangle the wires; there should be four "twisted pairs". Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.

Step 4 : Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the order. To do so, hold the plug with the clip side facing away from you and have the gold pins facing toward you as shown.

Step 5 : Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding, and if it isn't, that means that you stripped off too much shielding. simply snip the cables back a little more.

Step 6 : After the wires are securely sitting inside the plug, insert it into the crimping tool & push down.

⇒ It should be shaped correctly, but pushing too hard can crack the fragile plastic plug.

step 7 : Lastly, repeat for the other end using diagram B (to make a crossover cables) / using diagram A (to make a straight through cable).

⇒ To test it, plug it it and attempt to connect two devices directly.

Result:

Thus the different types of network cables are studied.